CHEMICAL REACTION NETWORKS
2. Non Mass-Action Kinetics
Michaelis-Menten and other kinetic models

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Other kinetic models of chemical reactions

In some cases the mass action kinetics could be considered too "low-level", and introduce unnecesary details in the models of some phenomena.

Examples of more abstract kinetic models are:

- the Michaelis-Menten kinetics (and Hill kinetics) of enzyme-catalysed reactions
- the logistic function for autocatalytic reactions

These kinetic models are approximations of the mass action kinetics that could be considered reasonable under certain conditions and that make models simpler or simlplify fitting real data

THE MICHAELIS-MENTEN KINETICS

The Michaelis-Menten kinetics

The Michaelis-Menten kinetics describes the dynamics of irreversible enzyme-catalysed reactions such as:

$$E + S \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} ES \xrightarrow{k_2} E + P$$

The assumptions under which the kinetics is an acceptable approximation of the mass action kinetics are the following:

- the whole enzyme-catalysed reaction is irreversible, namely the reaction $E + P \xrightarrow{k_{-2}} ES$ has a very small kinetic constant k_{-2} and can be ignored
- reactions $E + S \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} ES$ are very fast. Hence, the concentration of ES can be considered always in dynamic equilibrium.

The Michaelis-Menten kinetics

$$E + S \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} ES \xrightarrow{k_2} E + P$$

- by the dynamic equilibrium assumption we have $k_1[E][S] = (k_{-1} + k_2)[ES]$
- since the enzyme is not consumed we have $[E]_{tot} = [E] + [ES]$ constant
- hence, $k_1([E]_{tot} - [ES])[S] = (k_{-1} + k_2)[ES]$
- hence, $k_1[E]_{tot}[S] = (k_{-1} + k_2)[ES] + k_1[ES][S]$
- hence, $[E]_{tot}[S] = \frac{k_{-1}+k_2}{k_1}[ES] + [ES][S] = [ES](\frac{k_{-1}+k_2}{k_1} + [S])$
- hence, $[ES] = \frac{[E]_{tot}[S]}{\left(\frac{k_{-1}+k_2}{k_1}+[S]\right)}$ is the concentration of the complex ES at the dynamic equilibrium
- let $K_M = \frac{k_{-1}+k_2}{k_1}$, we obtain $[ES] = \frac{[E]_{tot}[S]}{K_M+[S]}$
- we know that $\frac{d[P]}{dt} = k_2[ES]$, hence, $\frac{d[P]}{dt} = \frac{k_2[E]_{tot}[S]}{K_M+[S]}$
- let $V_{max} = k_2[E]_{tot}$, we finally obtain:

$$\frac{d[P]}{dt} = \frac{V_{max}[S]}{K_M + [S]}$$

that is the Michaelis-Menten equation.

The Michaelis-Menten kinetics

In the Michaelis-Menten equation

$$\frac{d[P]}{dt} = \frac{V_{max}[S]}{K_M + [S]}$$

we have only two parameters:

- K_M (called Michaelis-Menten constant) that corresponds to $\frac{k_{-1}+k_2}{k_1}$
- V_{max} that corresponds to $k_2[E]_{tot}$ and is the max speed of production of P

The concentrations of the enzyme $[E]$ and $[ES]$ are no longer necessary variables in the ODE model of the considered reactions

The Michaelis-Menten kinetics

$$E + S \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} ES \xrightarrow{k_2} E + P$$

ODEs obtained from mass action kinetics:

$$\begin{cases} \frac{d[E]}{dt} = k_{-1}[ES] + k_2[ES] - k_1[E][S] \\ \frac{d[ES]}{dt} = k_1[E][S] - k_{-1}[ES] - k_2[ES] \\ \frac{d[S]}{dt} = k_{-1}[ES] - k_1[E][S] \\ \frac{d[P]}{dt} = k_2[ES] \end{cases}$$

ODEs obtained from Michaelis-Menten kinetics:

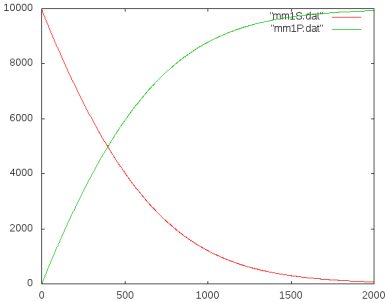
$$\begin{cases} \frac{d[S]}{dt} = -\frac{V_{max}[S]}{K_M+[S]} \\ \frac{d[P]}{dt} = \frac{V_{max}[S]}{K_M+[S]} \end{cases}$$

The Michaelis-Menten kinetics

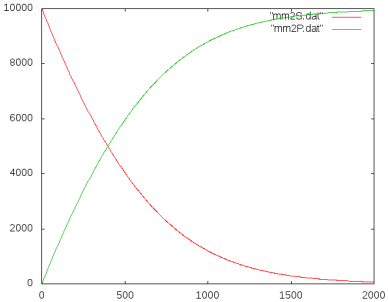
$$E + S \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} ES \xrightarrow{k_2} E + P$$

Numerical simulation with $k_1 = 0.1, k_{-1} = 1000, k_2 = 0.3$ and with initial concentrations $[E]_0 = 100, [S]_0 = 10000, [ES]_0 = [P]_0 = 0$:

Mass Action



Michaelis-Menten



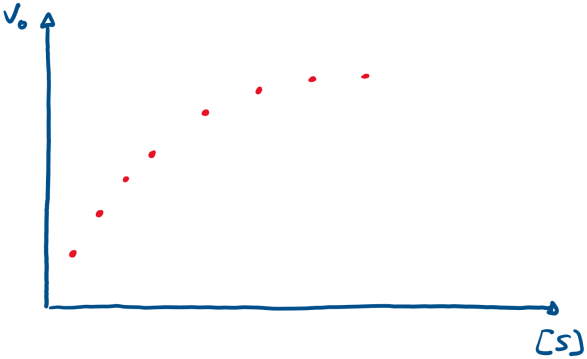
Experimental measurement of K_M and V_{max}

The parameters K_M and V_{max} are practical to determine in wet-lab experiments.

- Initial-rate method: measure the initial velocity v_0 for several substrate concentrations $[S]$ and fit the data to the Michaelis–Menten law

Example data (where v_0 is the initial rate of product P formation):

$[S]$ (Mol)	v_0 (Mol/s)
5	0.048
10	0.083
20	0.143
50	0.208
100	0.250
200	0.286
500	0.313



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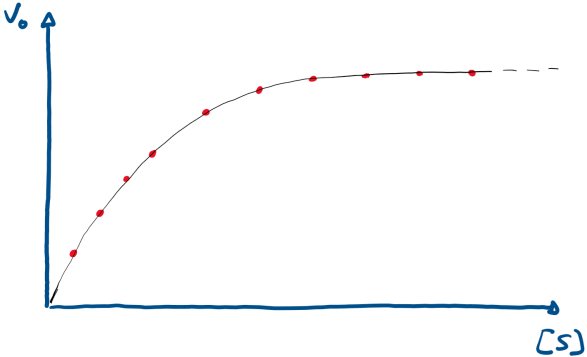

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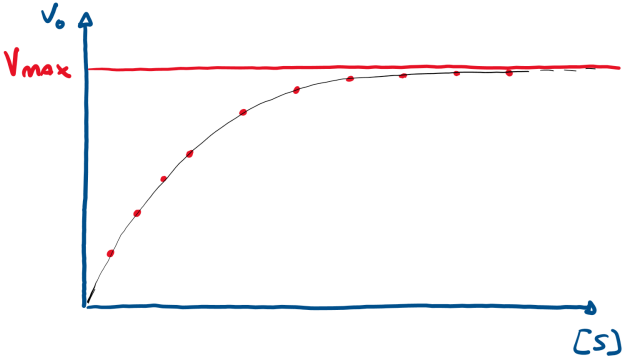

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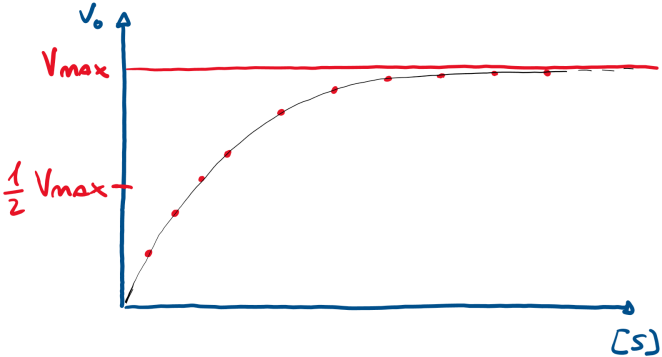

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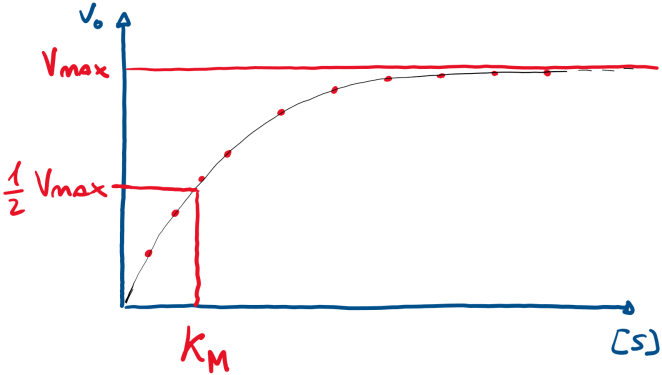


Interpretation of V_{max}

In the Michaelis–Menten equation

$$v = \frac{V_{max}[S]}{K_M + [S]},$$

the parameter V_{max} represents the maximum reaction rate attainable by the enzyme system.

Physical interpretation

- It corresponds to the rate achieved when all enzyme active sites are occupied:

$$[ES] = [E]_{tot}.$$

- In that case, $v = k_2[E]_{tot} = V_{max}$.
- Graphically, V_{max} is the horizontal asymptote of the v vs. $[S]$ curve.

Interpretation of K_M

Starting from the Michaelis–Menten equation:

$$v = \frac{V_{max}[S]}{K_M + [S]},$$

if we set $[S] = K_M$, we obtain:

$$v = \frac{V_{max} K_M}{K_M + K_M} = \frac{V_{max}}{2}.$$

Interpretation

- K_M is the substrate concentration at which the reaction rate is half of its maximum value.
- It provides a measure of the enzyme's affinity for the substrate:
  - smaller $K_M \rightarrow$ higher affinity (half-saturation at low $[S]$);
  - larger $K_M \rightarrow$ lower affinity.

THE HILL KINETICS

The Hill kinetics

The Hill kinetics is a generalization of Michaelis-Menten kinetics. It describes the dynamics of irreversible enzyme-catalysed reactions with cooperative binding such as:

$$\frac{d[P]}{dt} = \frac{V_{max}[S]^h}{K_M^h + [S]^h}$$

where h is the Hill coefficient and K is the half-saturation constant (the same as in Michaelis-Menten).

Cooperative binding: intuition and examples

Cooperative binding means that binding of one substrate molecule to a multimeric enzyme changes the probability that additional substrate molecules will bind to other sites of the same enzyme.

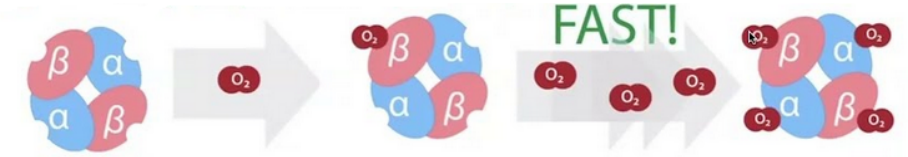
- Positive cooperativity ($h > 1$): the first binding event increases affinity of remaining sites (binding becomes easier); e.g. haemoglobin (oxygen binding).
- Negative cooperativity ($h < 1$): the first binding reduces affinity of remaining sites.
- No cooperativity ($h = 1$): independent binding sites; the Hill function reduces to standard Michaelis–Menten-like behaviour.

Mechanistic note: cooperativity often arises from conformational changes (T/R states) that propagate between subunits; the Hill coefficient h is an empirical measure of steepness (effective cooperativity), not necessarily the number of binding sites.

Cooperative binding: intuition and examples

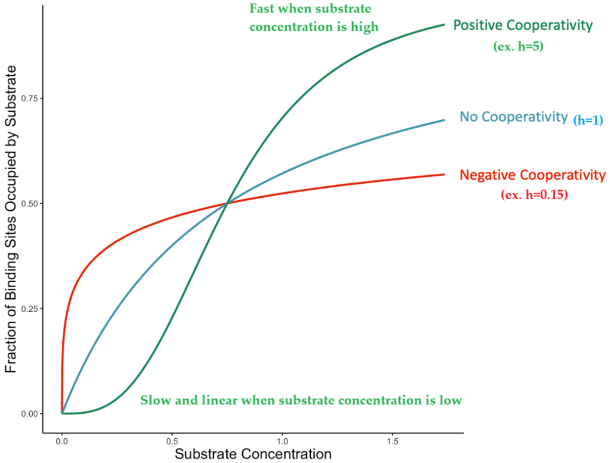
Example of cooperative binding:

- Haemoglobin: a tetrameric protein that transports oxygen in the blood. Binding of one oxygen molecule increases the affinity for additional oxygen molecules (positive cooperativity, $h \approx 2.8$).



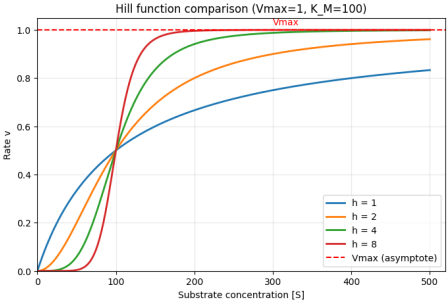
The Hill kinetics

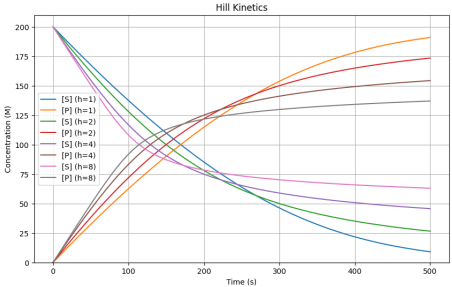
The intuition is that binding of one substrate by the enzyme favors (when $h > 1$) or disfavors (when $h < 1$) binding of additional substrates



Hill function behaviour: numeric example ($V_{max} = 1$, $K_M = 100$)

Compare the rate $v = \dfrac{V_{max}[S]^h}{K_M^h + [S]^h}$ for different Hill coefficients. Let $V_{max} = 1$ and $K_M = 100$.





Observations:

- For higher h the response is more switch-like (ultrasensitive).
- All curves still approach the same horizontal asymptote at $v = V_{max}$.
- The Hill coefficient is useful fit experimental data (fit h as an additional parameter).

THE LOGISTIC FUNCTION

The logistic function

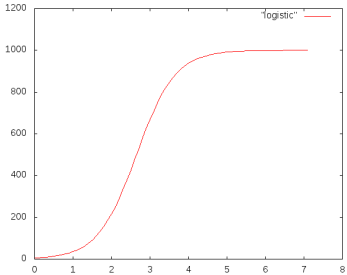
The logistic funciton finds common application in ecology to describe population growth in an enviroment with limited resources

A differential equation in which the logistic function is used is as follows:

$$\frac{dP}{dt} = \alpha P(1 - \frac{P}{K})$$

where α is a constant representing the growth rate and K is the carrying capacity, namely the maximal sustainable value of P

For example, with $\alpha = 2$ and $K = 1000$, starting from $P = 5$ we obtain the following dynamics:



The logistic function

In general, the logistic function can be used to describe the dynamics of something subject to an exponential growth combined with a saturation effect

An example of such kind of dynamics in (bio)chemistry are autocatalytic reactions, namely reactions such as

$$A + B \xrightarrow{k} 2B$$

where the product B catalyzes its own production

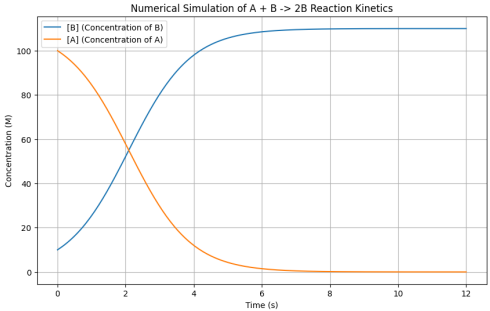
The logistic function and autocatalytic reactions

$$A + B \xrightarrow{k} 2B$$

ODEs obtained from mass action kinetics:

$$\begin{cases} \frac{d[A]}{dt} = -k_1[A][B] \\ \frac{d[B]}{dt} = k_1[A][B] \end{cases}$$

With $k = 0.01$, $[A]_0 = 100$ and $[B]_0 = 10$:



Let's make a few obsevations:

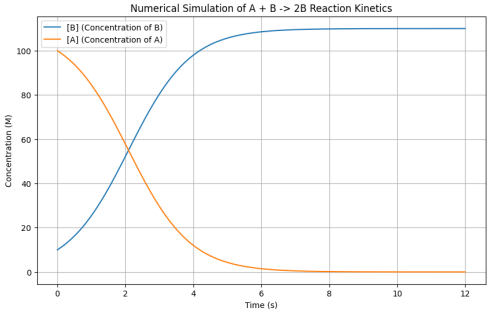
- the reaction transforms a into b, hence $T = [A] + [B] = [A]_0 + [B]_0$ is constant
- we have $\frac{d[B]}{dt} = k_1[A][B] = k_1(T - [B])[B] = k_1[B](T - [B]) = k_1 T[B](1 - \frac{[B]}{T})$
- this is a logistic function with $\alpha = k_1 T$ and $K = T$

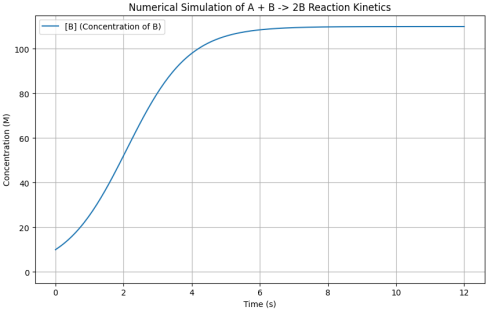
The logistic function

Numerical simulation with $k = 0.01$, $[A]_0 = 100$ and $[B]_0 = 10$ and $T = [A]_0 + [B]_0 = 110$:

ODEs obtained from mass action kinetics:

$$\begin{cases} \frac{d[A]}{dt} = -k_1[A][B] \\ \frac{d[B]}{dt} = k_1[A][B] \end{cases}$$



ODE obtained from logistic function:

$$\frac{d[B]}{dt} = k_1 T[B](1 - \frac{[B]}{T})$$